Modern Reinforcement Learning: From Agents to LLMs

Learning from Preferences

Lecture 7

Keywords: Preferences · Reward modeling · Pairwise comparisons · Bradley–Terry · Learned rewards

Ilija Bogunovic

Lecture 7: Learning from Preferences & Reward Modeling

- Learning from Preferences
  - Reward modeling
  - Pairwise comparisons
  - Bradley–Terry models
  - Learned rewards in RL
  - From control to alignment

📘 Side note / “Keep in mind” box:
— Rewards are not ground truth — they are models learned from data.

Why rewards are hard to specify

- Why rewards are hard
  - Proxy objectives
  - Misspecification
  - Reward hacking (https://openai.com/index/faulty-reward-functions/)
  - Human values ≠ scalar functions

Preferences as Indirect Supervision

- Preferences as supervision
  - Relative judgments
  - Easier than scoring
  - Partial information

📘 Side note / “Keep in mind” box:
— Preferences tell us what is better, not how good something is.

Where Preference Learning Fits in RL

- Learning objectives from data
  - Behavior data vs preference data
  - Reward learning vs policy learning
  - Offline and online settings
  - Two-stage learning pipeline

What Is Preference Feedback?

- Preference feedback
  - Comparisons between trajectories
  - $\tau_i \succ \tau_j$ indicates preferences
  - No absolute reward values
  - Possibly noisy and inconsistent

Preference Learning Problem Setup

- Preference learning setup
  - MDP without known reward
  - Trajectories $\tau = (s_0, a_0, \ldots, s_T)$
  - Latent reward $r_\theta(s, a)$
  - Preference data $\mathcal{D} = \{(\tau_i \succ \tau_j)\}$

📘 Side note / “Keep in mind” box:
— The reward is a latent variable, preferences are the observations.

Bradley–Terry Preference Model

Preference model (Bradley–Terry):

- Trajectory return

- Pairwise preference likelihood

- Logistic noise assumption

- Only differences in return are identifiable

Log-Likelihood of Preference Data

Preference learning objective

- Preference dataset

- Probabilistic model

Identifiability and Invariances

- Identifiability of reward models
- Additive invariance

$$r_\theta(s,a) \;\mapsto\; r_\theta(s,a) + c \quad \Rightarrow \quad \mathbb{P}(\tau_i \succ \tau_j) \text{ unchanged}$$

- Return differences are invariant

$$R_\theta(\tau_i) - R_\theta(\tau_j)$$

- Absolute reward scale is unidentifiable
- Multiple reward functions explain same preference

📘 Side note / "Keep in mind" box:
—Reward learning identifies an equivalence class of rewards, not a unique function.

Gradient and Optimization Properties

Optimization of preference objective

- Gradient of log-likelihood

$$\nabla_\theta \ell(\theta) = -\sum_{(i,j)\in\mathcal{D}} \left(y_{ij} - \sigma(\Delta R_{ij})\right) \nabla_\theta \Delta R_{ij}$$

where

$$\Delta R_{ij} := R_\theta(\tau_i) - R_\theta(\tau_j)$$

- Linear reward ⇒ convex objective

- Neural reward ⇒ non-convex optimization

Structure of the Learned Reward

What the reward model learns

- Return decomposition

$$R_\theta(\tau) = \sum_{t=0}^{T} r_\theta(s_t, a_t)$$

- Gradient contribution

$$\nabla_\theta \Delta R_{ij} = \sum_t \nabla_\theta r_\theta(s_t^i, a_t^i) - \sum_t \nabla_\theta r_\theta(s_t^j, a_t^j)$$

- Credit assignment across trajectories

- No supervision at single time steps

Structure of the Learned Reward

Generalization limits

- Preferences collected under policy π_{data}

- Reward evaluated under policy π_θ

- Off-policy reward learning

- Extrapolation beyond observed trajectories

Optimization Mismatch with Learned Rewards

Objective mismatch

- True (unknown) objective

$$J^\star(\pi) := \mathbb{E}_{\tau\sim\pi}\left[R^\star(\tau)\right]$$

- Learned reward objective

$$J_\theta(\pi) := \mathbb{E}_{\tau\sim\pi}\left[R_\theta(\tau)\right]$$

- Policy optimization step

$$\pi_{k+1} \in \arg\max_{\pi} J_\theta(\pi)$$

- No guarantee:

$$J^\star(\pi_{k+1}) \geq J^\star(\pi_k)$$

Error Amplification Under Policy Optimization

Error amplification

- Reward model error

$$\varepsilon(\tau) := R_\theta(\tau) - R^\star(\tau)$$

- Decomposition of learned objective

$$J_\theta(\pi) = J^\star(\pi) + \mathbb{E}_{\tau\sim\pi}\left[\varepsilon(\tau)\right]$$

- Policy improvement step

$$\pi_{k+1} \in \arg\max_\pi \left(J^\star(\pi) + \mathbb{E}_{\tau\sim\pi}[\varepsilon(\tau)] \right)$$

- Bias toward trajectories with large $\varepsilon(\tau)$

Modern Parallels: Reward Models & RLHF

Modern Parallels: Reward Models & RLHF

Preference learning for LLMs

- Prompt–response pairs (x, y)
- Human preference data $(x, y_i \succ y_j)$
- Reward model $R_\theta(x, y) \in \mathbb{R}$
- Trained via Bradley–Terry objective

$$\mathbb{P}(y_i \succ y_j \mid x) = \sigma(R_\theta(x, y_i) - R_\theta(x, y_j))$$

📘 Side note / “Keep in mind” box:
— Reward models for LLMs are trajectory-level rewards over token sequences.

RLHF as Two-Stage Optimization

Decomposition of RLHF

- Stage 1: Reward learning

Preference data: $(x, y_i \succ y_j) \;\Rightarrow\; R_\theta(x, y)$

- Stage 2: Policy optimization

Language model policy $\pi_\phi(y \mid x)$

$$\phi^\star \in \arg\max_\phi \mathbb{E}_{y \sim \pi_\phi(\cdot|x)}\left[R_\theta(x, y)\right]$$

Scalarization and Value Collapse

- Limits of scalar reward models
- Human preferences are multi-dimensional $v(x, y) \in \mathbb{R}^k$
- Reward model learns scalarization $R_\theta(x, y) \approx w^\top v(x, y)$
- Weights w are implicit and data-dependent
- Different preferences ⇒ same scalar reward

📘 Side note / “Keep in mind” box:
— Scalar rewards hide trade-offs rather than resolving them.

Failure Modes in Practice

- Observed issues in RLHF systems
  - Over-optimization of R_θ
  - Distribution shift: $(x, y)_{\text{train}} \neq (x, y)_{\text{deploy}}$
  - Reward hacking in language space
  - Sensitivity to prompt and decoding changes

Demo

Preference Learning in a Gridworld

Setup

- Gridworld MDP $(\mathcal{S}, \mathcal{A}, P, \gamma)$, unknown $r^\star$

- Trajectories $\tau = (s_0, \ldots, s_T)$

- Simulated preferences: $\mathbb{P}(\tau_i \succ \tau_j) = \sigma\Big(R^\star(\tau_i) - R^\star(\tau_j)\Big)$

- Learn $r_\theta(s)$ from comparisons:

$$\theta^\star = \arg\min_\theta -\sum_{(i,j)} \log \sigma(R_\theta(\tau_i) - R_\theta(\tau_j))$$

Reward Heatmaps and Policy Change

Outcome

- Learned reward close to true reward
- Heatmaps differ slightly
- Optimal paths differ: $\pi^\star_{r^\star} \neq \pi^\star_{r_\theta}$
- Takeaway: small reward errors → large behavioral changes